Bachelor of Information Technology
BDS 23114 Data Analytics

Loh Min Yu 202410034
Fergusson Chim Yuan Liang 202430223
Kelvin Chia Kai Vun BDS_A2201F-2501001

FOOD WASTE ANALYTICS AT MAMAK EATERIES

OBJECTIVE

To analyze how customer volume and food preparation volume affect food waste. Identifying waste patterns, provide recommendations for improvement

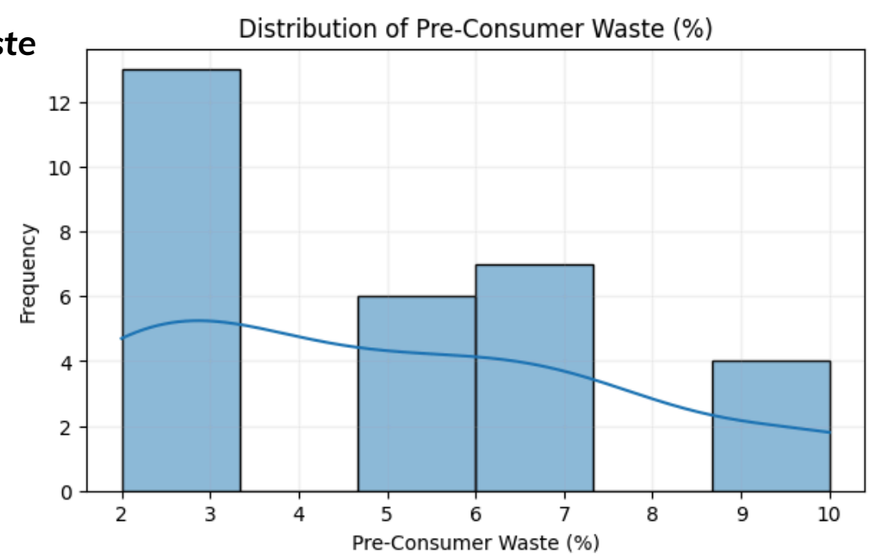
KEY FINDING

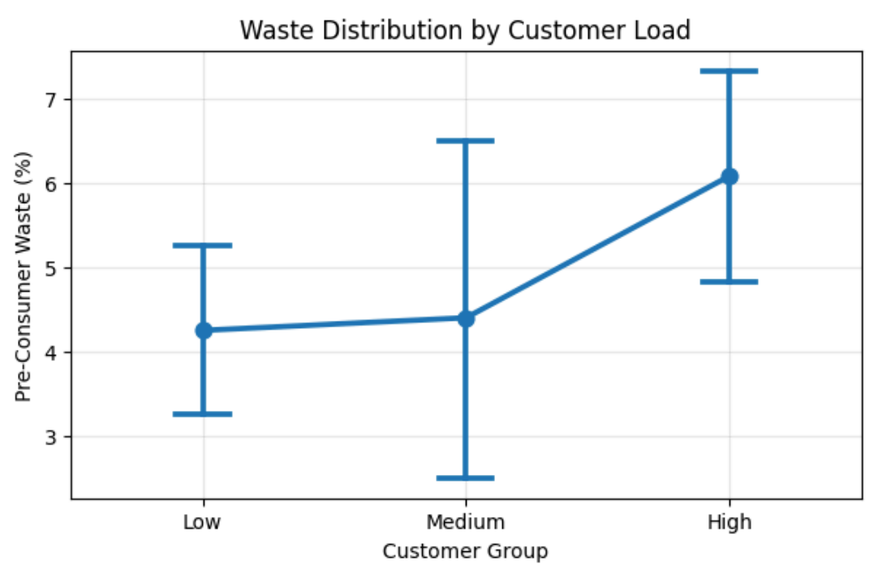
- Shops keep wastage rate low
- High customer traffic; higher waste
- More serving per day; lesser wastage
- Customer volume ≠ serving prep

VISUALIZATION

Histogram of Pre-Consumer Waste

Food waste in most stores is concentrated within the range of 3%–4%, indicating that overall waste is controlled within a reasonable range, but some stores still have relatively high levels of waste.





Waste Distribution by Customer Load

The higher the passenger flow, the higher the waste rate. The stocking strategy during high passenger flow periods needs to be optimized, otherwise it will lead to a significant increase in waste.

METHODOLOGY

- Interviewing
- Dataset Processing & Visualization
- Python Coding

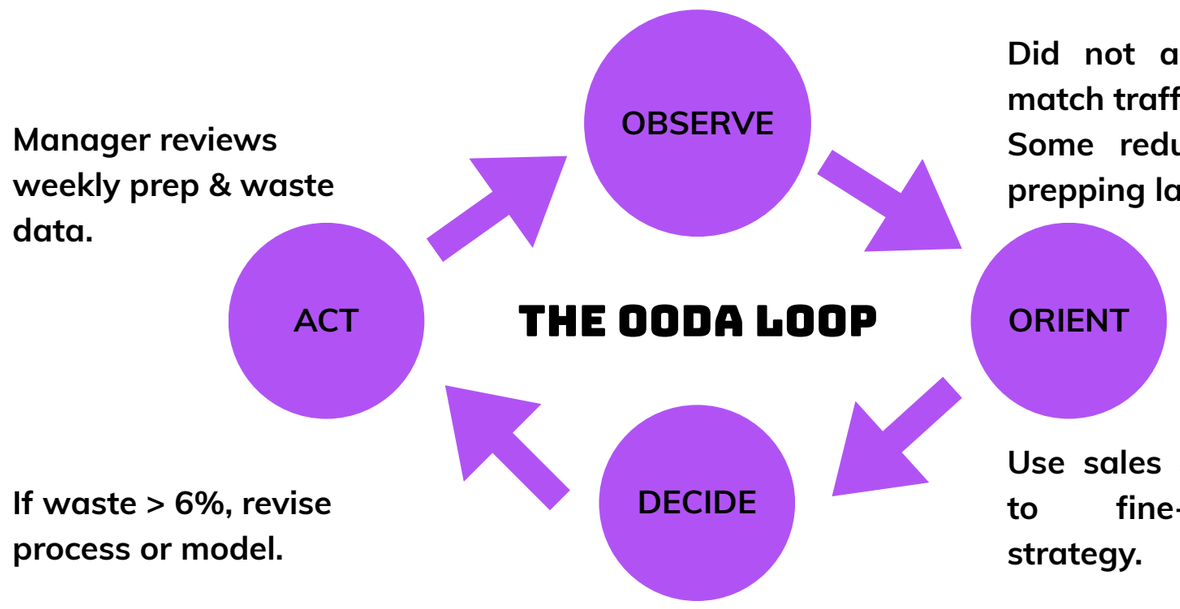
Correlation Heatmap

There is a positive correlation between daily customer numbers and waste rate (0.23), while "supply per item" is negatively correlated with waste (-0.34), indicating that adjusting the supply per item can effectively reduce waste.

RECOMMENDATION

- Determine portion sizes based on solid traffic forecasts
- Combine portions to create popular dishes
- Set waste targets based on customer demographics
- Implement preparation records and prediction adjustments

Waste mainly from unused ingredients & overcooking.
High-traffic days: waste can reach 10%.
Normal days: 2–3% waste.



Manager reviews weekly prep & waste data.

Did not adjust prep to match traffic.
Some reduce waste by prepping larger portions.

Use sales & waste data to fine-tune prep strategy.

If waste > 6%, revise process or model.

Forecast prep needs by traffic levels: (high / med / low)
Consider: traffic, prep quantity, sales, waste %.